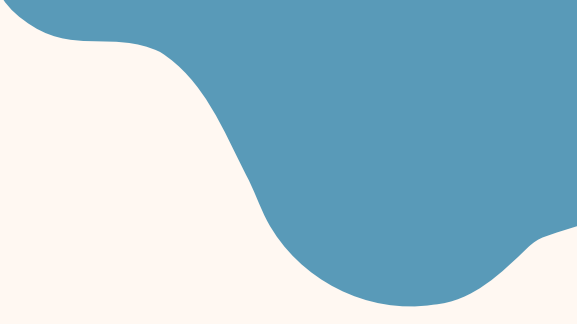




UNDERSTANDING PRESSURE: ATMOSPHERIC AND FLUID

A Fun STEM Adventure for Grade 6

WHAT IS PRESSURE?

Pressure is the force pushing on an area. Think of it like this – when you press your hand on a table, you're creating pressure!

- Pressure depends on how much force is applied
- Pressure can happen in air, water, and other fluids

The more force you use, the more pressure you create!



MEET ATMOSPHERIC PRESSURE

PRESSURE FROM THE AIR ALL AROUND US

Did you know that air has weight? The atmosphere surrounding Earth is made up of tiny air particles that press down on everything, including us! This invisible force pushing on us from all directions is called atmospheric pressure.



HOW DOES ATMOSPHERIC PRESSURE WORK?

IT PRESSES DOWN ON US

Air has weight and pushes down on us and everything around us, like an invisible blanket covering the Earth.

WE DON'T FEEL IT

We don't notice the pressure because it pushes equally from all directions, balancing out perfectly!

IT AFFECTS WEATHER & BREATHING

Changes in atmospheric pressure can bring rain or sunshine, and even affect how easily we breathe.



EXPERIMENTS WITH ATMOSPHERIC PRESSURE



Try these fun experiments at home! Crushing Can
Demo: Heat an empty can, then quickly flip it
into cold water – watch it crush! Inverted Glass
Trick: Fill a glass with water, cover with a card,
and flip it upside down – the water stays inside!



WHAT IS FLUID PRESSURE?

DEFINITION

Fluid pressure is the pressure inside a liquid or gas that pushes in all directions at the same time.

KEY POINTS TO REMEMBER

Fluid pressure pushes equally in all directions.
It increases with depth in liquids.





FLUID PRESSURE IN EVERYDAY LIFE

1. DAM POWER

Water pushes against dam walls with tremendous force, helping generate electricity!



1. TIRE PRESSURE

Air pressure inside tires keeps them firm and helps vehicles move smoothly.

1. BLOOD FLOW

Blood pressure pushes blood through our veins, delivering oxygen to every part of our body.



EXPERIMENT: WATER PRESSURE DEMO

Try this fun experiment at home!

1. Take a plastic bottle and poke 3 holes at different heights
2. Cover the holes with tape and fill with water
3. Remove the tape and watch what happens!

Notice: Water shoots out faster from the bottom hole because pressure increases with depth!

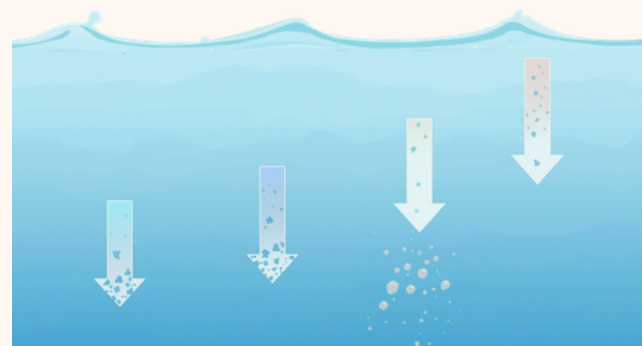


WHY DOES PRESSURE INCREASE WITH DEPTH?

At the top of the water:

- Less water above you
- Less weight pushing down
- Lower pressure on your body

Think of it like a stack of books – one book is light!



THE SCIENCE BEHIND IT

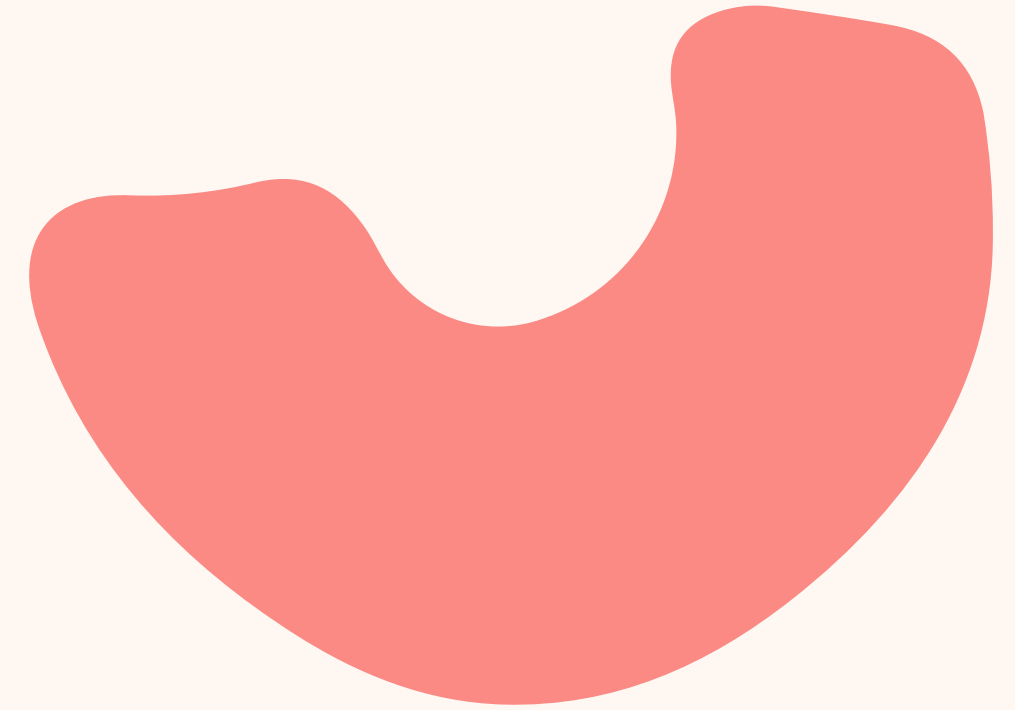
At the bottom of the water:

- More water above you
- More weight pushing down
- Higher pressure all around

More books stacked = heavier load!

ATMOSPHERIC PRESSURE

- Pushes down from the air above us
- Created by the weight of air in our atmosphere
- Affects weather patterns and storms
- Same pressure all around us at sea level
- Measured using a barometer



FLUID PRESSURE

- Pushes equally in all directions
- Found inside liquids and gases
- Increases with depth underwater
- More fluid above = more pressure below



WHY IS PRESSURE IMPORTANT?



UNDERSTANDING WEATHER PATTERNS

Atmospheric pressure changes help us predict rain, storms, and sunny days. Weather forecasters use pressure readings every day!

SUBMARINES & BALLOONS

Submarines use pressure to dive deep underwater. Hot air balloons rise because warm air creates different pressure. Science makes them work!

OUR BODY FUNCTIONS

Blood pressure keeps our heart pumping blood through our body. Pressure helps us breathe air into our lungs every moment!



QUICK RECAP: KEY POINTS

1. WHAT IS PRESSURE?

Pressure is the force pushing on an area. The more force applied, the greater the pressure!



1. ATMOSPHERIC PRESSURE

Comes from the weight of air all around us. Air has weight and pushes on everything!

1. FLUID PRESSURE

Pushes in all directions inside liquids and gases. Pressure increases as you go deeper!



FUN QUIZ TIME!

1. What pushes on us from the air?
2. Does fluid pressure push equally in all directions?

1. What happens to water pressure as you go deeper?
2. Name one example of atmospheric pressure in action.



SUMMARY AND WHAT'S NEXT?

You've learned so much about pressure today! Now it's time to explore the world around you.

- Observe pressure effects in everyday life
- Try simple experiments at home or school
- Keep asking questions and exploring STEM topics!

Pressure is everywhere – from the air above us to the water in our bottles. Keep discovering!





**THANK YOU FOR
LEARNING
ABOUT PRESSURE
WITH US!**

ADDITIONAL RESOURCES

Keep exploring and learning more about pressure! Here are some helpful resources to continue your STEM adventure.

